



Lecture 2 — Before class

Introduction to Quantitative Biology | Read before Thursday

In Lecture 1 we built the simplest model of a gene, $dY/dt = \beta - \alpha Y$, and solved it. In it, β was a fixed number: the gene was simply on.

Lecture 2 asks what happens when a gene is **regulated** — when its production rate depends on the concentration of another protein — and then looks at how those regulatory links are arranged across a whole genome. Three things to do before Thursday.

1. **Work through Section 1**, where we derive the production rate of a gene switched on by an activator. It is one page of algebra and you should do it with a pen. In class we do the same calculation for a *repressor*, and it will go quickly if this one is fresh.
2. **Read Chapter 2 of Uri Alon's book [1]** (Section 2 tells you which parts matter most).
3. **Write down your answers to the questions in Section 3**, and bring them. We start the lecture from them.

As always, a precise question is as useful as an answer. All PDFs are on the GitHub page of the course: <https://github.com/qbiomaster/qBioMaster-introduction>.

Conventions used in this course

We follow the notation of Alon's book [1], and it is worth stating it once.

- **A symbol standing for a molecule means its concentration.** X is the concentration of the transcription factor X , D the concentration of free binding sites, Y the concentration (or the number per cell — we will not need to distinguish) of the protein a gene makes. There is no separate notation for “the molecule” and “how much of it there is”.
- **Square brackets appear only on names made of more than one letter**, such as the complex $[XD]$. They are there to say that XD is *one species*, the site with X bound to it — not the product of X and D . Without them, XD in an equation would be read as a multiplication, which is exactly what it is not.
- β is a **production rate** and α a **removal rate**, as in Lecture 1, where $dY/dt = \beta - \alpha Y$.
- K is a concentration; n , the Hill coefficient, is a pure number.

What we will do on Thursday

- A short recap of the three ways we now have of describing the same gene — the differential equation, a simulation of single cells, and the distribution over cells — and the answer to a question we left open: *why* that distribution is a Poisson.
- The **input function**: how the production rate of a gene depends on the transcription factor that regulates it. You derive the activator case below; we do the repressor together.
- **Network motifs**: taking the whole transcription network of *E. coli* and asking which patterns in it occur far more often than chance would put there. We start with the simplest, a gene that regulates itself, and ask what it is good for.

1. The input function of a gene

A transcription factor is a protein that binds a specific site in the promoter of a gene and changes how often that gene is transcribed. If X is its concentration, the production rate of the regulated protein is no longer a constant β but a function $f(X)$, and the equation from Lecture 1 becomes

$$\frac{dY}{dt} = f(X) - \alpha Y.$$

Our job in this section is to find f for an **activator**: a factor whose binding turns transcription *on*.

Step 1: the site is either free or occupied

The promoter has one binding site for X . Call D the concentration of sites that are free and $[XD]$ the concentration of sites with X bound to them. Every site is in one of those two states, so

$$D + [XD] = D_T, \quad (1)$$

where D_T is the total concentration of sites. (For a single site in a bacterium, $D_T \approx 1$ molecule per cell volume.)

Step 2: binding and unbinding

X and D diffuse, collide, and stick together. The rate at which complexes form is proportional to how much of each is around, $k_{\text{on}}XD$; and each complex falls apart on its own at rate k_{off} , giving $k_{\text{off}}[XD]$ in total. So

$$\frac{d[XD]}{dt} = k_{\text{on}}XD - k_{\text{off}}[XD]. \quad (2)$$

This is the same kind of equation as Lecture 1

Equation (2) has a production term and a removal term, exactly like $dY/dt = \beta - \alpha Y$. We are going to do to it what we did there: look for the steady state, where the left-hand side is zero.

Step 3: the binding equilibrium — and why we may use it

Here is the step that makes everything else easy, and it is worth pausing on.

Binding and unbinding are **fast**: a transcription factor finds its site, or lets go of it, on a timescale of seconds. The concentration of a protein, on the other hand, changes on the timescale of the cell cycle — tens of minutes to hours. So on the timescale we care about, the site has *already* reached whatever balance it is going to reach, thousands of times over.

We may therefore set $d[XD]/dt = 0$ in Equation (2). That does not mean binding has stopped: it means binding and unbinding have come into balance, each still going on at the same rate as the other, and that they reached that balance long before X had time to change. This gives

$$k_{\text{on}}XD = k_{\text{off}}[XD] \quad \implies \quad [XD] = \frac{XD}{K}, \quad \text{where} \quad K = \frac{k_{\text{off}}}{k_{\text{on}}}.$$

K is called the **dissociation constant**. It has units of concentration, and the bigger it is, the more readily the complex falls apart — that is, the *weaker* the binding.

Step 4: how often is the site occupied?

Put $[XD] = XD/K$ into the conservation law (1):

$$D + \frac{X}{K}D = D_T \quad \Rightarrow \quad D \left(1 + \frac{X}{K}\right) = D_T \quad \Rightarrow \quad D = \frac{D_T}{1 + X/K}.$$

The occupied sites are what is left over:

$$[XD] = D_T - D = D_T \left(1 - \frac{1}{1 + X/K}\right) = D_T \frac{X/K}{1 + X/K} = D_T \frac{X}{K + X}.$$

Divide by D_T and you have the quantity we were after — the **fraction of the time the site is occupied**, a pure number:

$$\frac{[XD]}{D_T} = \frac{X}{K + X}.$$

Step 5: from occupancy to the production rate

An activator works by being bound: transcription happens when X sits on the site. So the production rate is proportional to how often the site is occupied, and if β is the rate when the site is occupied all the time,

$$f(X) = \beta \frac{X}{K + X} \quad (3)$$

KEY TAKEAWAY

Read the formula in three places and it tells you what its two parameters mean.

- $f(0) = 0$: **no activator, no transcription.**
- $f(K) = \beta/2$: **K is the concentration at which the gene runs at half speed. This is what makes K measurable rather than abstract.**
- $f(X) \rightarrow \beta$ when $X \gg K$: **saturation. Once the site is occupied essentially all the time, adding more activator changes nothing.**

Step 6: several molecules at once — the Hill function

Very often a transcription factor does not act alone: n copies must bind together before the promoter responds. The algebra is the same, with X^n in place of X , and the result is the **Hill function**

$$f(X) = \beta \frac{X^n}{K^n + X^n}, \quad (4)$$

where n is the **Hill coefficient**. Equation (3) is the case $n = 1$. Note that $f(K) = \beta/2$ whatever n is, so K keeps its meaning; what n changes is how *sharply* the gene switches, which is the subject of Exercise (c) below.

Exercise 1: Check yourself

Solutions are at the end of this sheet.

(a) From $K = k_{\text{off}}/k_{\text{on}}$, and knowing that k_{on} is a rate per unit concentration while k_{off} is a rate, check that K has units of concentration. Two promoters have $K = 1$ nM and $K = 100$ nM. Which one binds its activator more tightly?

(b) Verify directly from Equation (4) that $f(K) = \beta/2$ for every n .

(c) How sharp is the switch? Find the concentration X at which the gene reaches 90% of β , as a multiple of K , for $n = 1$, $n = 2$ and $n = 4$. Do the same for 10%. Then compute, for each n , the ratio between those two concentrations — the factor by which X must change to take the gene from nearly off to nearly on. What does a large n buy?

(d) Sketch $f(X)$ against X for $n = 1$ and $n = 4$ on the same axes, marking K and β . You do not need a computer; you need the three values from the takeaway box and your answer to (c).

Exercise 2: A question for Thursday

We derived the activator. Now think about a **repressor**, where X binding switches the gene *off*, so that transcription happens only while the site is **free**.

Nothing in Steps 1–4 assumed what the factor does once it is bound — those steps only worked out how often the site is occupied. So you already have everything you need.

Which of the two quantities from Step 4 is the relevant one now, and what does $f(X)$ become? Write down your guess, sketch it, and check the two limits ($X = 0$ and $X \gg K$) against what you expect a repressor to do. We derive it properly in class.

2. Reading Alon, Chapter 2

Chapter 2 of *An Introduction to Systems Biology* [1] is about **network motifs**. Read in this order of priority:

1. **Sections 2.1 and 2.2** (patterns, randomized networks, network motifs) — read carefully. The idea to take away is the method, not the biology: you cannot tell whether a pattern is meaningful until you know how often it would occur by accident.
2. **Section 2.3** (autoregulation is a network motif) — read carefully, pen in hand. It counts self-arrows in *E. coli* and compares that count with a random network. We redo that count on the real data file in class.
3. **Section 2.4** (negative autoregulation speeds the response time) — read. We derive this together on Thursday, so it is enough to follow the argument rather than reproduce it.
4. Section 2.5 (robustness) — optional.

The Hill function is in Chapter 1 as well

Sections 1.3.3 to 1.3.5 of Chapter 1 — input functions and Hill functions — were marked optional in the sheet for Lecture 1, with the note that we would come back to them. This is when. If you want a second account of Section 1 above, in the author's own words and with the activator and the repressor side by side, that is where to look.

If you want the details behind Section 1

The calculation above follows Appendix A of the same book: **A.1** for the binding equilibrium and **A.6** for the activator. The repressor, which we do in class, is **A.5**. The appendix is on the course page as well. It is not required reading — Section 1 of this sheet is self-contained — but it is where the details live.

3. Questions to bring on Thursday

Write your answers down. We start the lecture from them.

Exercise 3: Write down your answers

1. The repressor. Your guess from the exercise above: what is $f(X)$ for a repressor, and what does it do at $X = 0$ and at large X ?

2. Why compare with a random network? Alon does not simply report that *E. coli* has 40 genes regulating themselves. He first works out how many a randomized network would have. Why is the raw count, on its own, not an argument? Answer in two or three sentences.

3. A guess, before we measure it. A gene that represses itself reaches a lower steady state than the same gene without that repression. So to compare the two fairly we must give the self-repressing one a stronger promoter, until both settle at the same level.

Once we have done that, which of the two do you expect to *reach* that level faster — or do you expect no difference? There is no wrong answer here; the point is to have committed to one before we compute it.

Solutions to the “check yourself” exercises

(a) k_{on} has units of $1/(\text{time} \times \text{concentration})$ and k_{off} units of $1/\text{time}$, so their ratio $K = k_{\text{off}}/k_{\text{on}}$ has units of concentration. The promoter with $K = 1$ nM binds *more tightly*: a small K means the gene is already at half activity at a low concentration of activator, which is another way of saying the complex is reluctant to fall apart.

(b) Substituting $X = K$ into Equation (4): $f(K) = \beta K^n / (K^n + K^n) = \beta K^n / (2K^n) = \beta/2$, and the K^n cancels whatever n is.

(c) Setting $\beta X^n / (K^n + X^n) = 0.9\beta$ gives $X^n = 9K^n$, so $X = 9^{1/n}K$; the same steps with 0.1 give $X = (1/9)^{1/n}K$.

	$n = 1$	$n = 2$	$n = 4$
X at 10% of β	$0.111 K$	$0.333 K$	$0.577 K$
X at 90% of β	$9 K$	$3 K$	$1.732 K$
ratio between them	81	9	3

With $n = 1$ the activator concentration must change by a factor of **81** to take the gene from nearly off to nearly on; with $n = 4$, a factor of **3** is enough. A large Hill coefficient buys a **sharp switch**: the gene distinguishes between two nearby concentrations instead of responding gradually over two orders of magnitude.

(d) Both curves start at the origin, pass through $(K, \beta/2)$, and flatten towards β . The $n = 1$ curve bends immediately and approaches β slowly; the $n = 4$ curve stays low, rises steeply near $X = K$, and flattens quickly — close to a step at $X = K$.

References

- [1] Uri Alon. *An Introduction to Systems Biology: Design Principles of Biological Circuits*. 2nd. CRC Press, 2019.